

The Law of Cosines

Name
Date
Course

Solving a Triangle: Finding all three side lengths and all three interior angle measures of a triangle.

Scenarios when Solving Triangles

The Law of Cosines

SSS SAS

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = c^2 + a^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

The Law of Sines

AAS

ASA

SSA

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

↓
Ambiguity

i) No triangle

ii) One triangle

iii) Two triangles

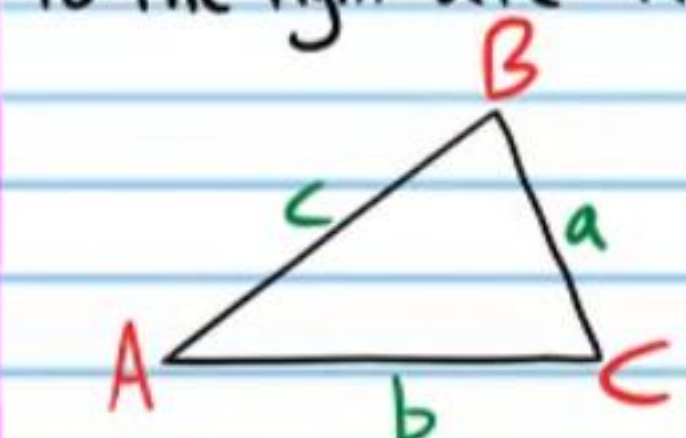
The Law of Cosines

If a , b , and c are the side lengths of a triangle, and A , B , and C are the interior angle measures, then the equations to the right are true.

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

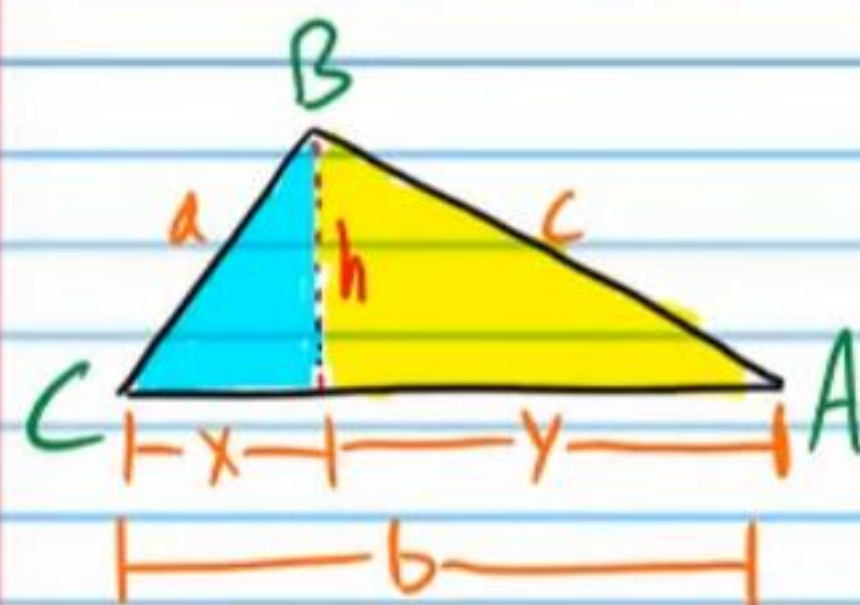
$$c^2 = a^2 + b^2 - 2ab \cos C$$



SSS or SAS

* The law of cosines can be used when all three side lengths are known, or when two sides lengths are known along with an angle measure between the two sides.

Proof of the Law of Cosines



$$\sin C = \frac{h}{a}$$

$$a \sin C = h$$

$$\cos C = \frac{x}{a}$$

$$a \cos C = x$$

Pythagorean Theorem

$$c^2 = h^2 + y^2$$

$$c^2 = (a \sin C)^2 + (b - x)^2$$

$$c^2 = a^2 \sin^2 C + (b - x)(b - x)$$

$$c^2 = a^2 \sin^2 C + b^2 - bx - bx + x^2$$

$$c^2 = a^2 \sin^2 C + b^2 - 2bx + x^2$$

$$c^2 = a^2 \sin^2 C + b^2 - 2ba \cos C + (a \cos C)^2$$

$$c^2 = a^2 \sin^2 C + b^2 - 2ba \cos C + a^2 \cos^2 C$$

$$c^2 = a^2 \sin^2 C + a^2 \cos^2 C + b^2 - 2ab \cos C$$

$$c^2 = a^2 (\sin^2 C + \cos^2 C) + b^2 - 2ab \cos C$$

$$c^2 = a^2 (1) + b^2 - 2ab \cos C$$

$$\boxed{c^2 = a^2 + b^2 - 2ab \cos C}$$

The Side-Angle-Side Scenario (SAS)

Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle.

① $B = 19^\circ$, $c = 10$ ft, $a = 14$ ft

I) Find b .

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$b^2 = 14^2 + 10^2 - 2(14)(10) \cos 19^\circ$$

$$b^2 = 196 + 100 - 280 \cos 19^\circ$$

$$b^2 \approx 296 - 264.745...$$

$$b^2 \approx 31.254...$$

$$\sqrt{b^2} \approx \pm \sqrt{31.254...}$$

$$b \approx \pm 5.590599$$

$$\boxed{b \approx 5.590599 \text{ ft}}$$

II) Find $m\angle C$.

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin C}{10} \approx \frac{\sin 19^\circ}{5.590599}$$

$$\sin C \approx \frac{10 \sin 19^\circ}{5.590599}$$

$$\sin C \approx 0.5823...$$

$$C \approx \sin^{-1}(0.5823...)$$

$$\boxed{C \approx 35.61595^\circ}$$

III) Find $m\angle A$.

$$A + B + C = 180^\circ$$

$$A + 19^\circ + 35.61595^\circ \approx 180^\circ$$

$$\boxed{A \approx 125.38^\circ}$$

② $b = 1.6$ in, $c = 3.4$ in, $A = 100^\circ$

I) Find a .

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$a^2 = 1.6^2 + 3.4^2 - 2(1.6)(3.4) \cos 100^\circ$$

$$a^2 = 2.56 + 11.56 - 10.88 \cos 100^\circ$$

$$a^2 \approx 14.12 - 10.88(-0.1736...)$$

$$a^2 \approx 14.12 + 1.889...$$

$$a^2 \approx 16.009...$$

$$\boxed{a \approx 4.00116... \text{ in}}$$

II) Find $m\angle B$.

$$\frac{\sin B}{b} = \frac{\sin A}{a}$$

$$\frac{\sin B}{1.6} \approx \frac{\sin 100^\circ}{4.00116...}$$

$$\sin B \approx \frac{1.6 \sin 100^\circ}{4.00116...}$$

$$\sin B \approx 0.3938...$$

$$B \approx \sin^{-1}(0.3938...)$$

$$\boxed{B \approx 23.1917^\circ}$$

III) Find $m\angle C$.

$$A + B + C = 180^\circ$$

$$100^\circ + 23.1917^\circ + C \approx 180^\circ$$

$$\boxed{C \approx 56.8^\circ}$$

③ $a = 7.2 \text{ cm}$, $C = 57^\circ$, $b = 8 \text{ cm}$

I) Find c .

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 7.2^2 + 8^2 - 2(7.2)(8) \cos 57^\circ$$

$$c^2 = 51.84 + 64 - 115.2 \cos 57^\circ$$

$$c^2 \approx 115.84 - 62.742...$$

$$c^2 \approx 53.09758...$$

$$\sqrt{c^2} \approx \sqrt{53.09758...}$$

$$\boxed{c \approx 7.2868 \text{ cm}}$$

II) Find $m\angle A$.

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin A}{7.2} \approx \frac{\sin 57^\circ}{7.2868}$$

$$\sin A \approx \frac{7.2 \sin 57^\circ}{7.2868}$$

$$\sin A \approx 0.82868...$$

$$A \approx \sin^{-1}(0.82868...)$$

$$\boxed{A \approx 55.9634^\circ}$$

III) Find $m\angle B$.

$$A + B + C = 180^\circ$$

$$55.9634^\circ + B + 57^\circ = 180^\circ$$

$$\boxed{B \approx 67^\circ}$$

• ④ $A = 38^\circ$, $b = 5.9 \text{ mm}$, $c = 12 \text{ mm}$

I) Find a .

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$a^2 = 5.9^2 + 12^2 - 2(5.9)(12) \cos 38^\circ$$

$$a^2 = 34.81 + 144 - 141.6 \cos 38^\circ$$

$$a^2 \approx 178.81 - 111.5823...$$

$$a^2 \approx 67.2276...$$

$$\sqrt{a^2} \approx \sqrt{67.2276...}$$

$$\boxed{a \approx 8.1992 \text{ mm}}$$

II) Find $m\angle B$.

$$\frac{\sin B}{b} = \frac{\sin A}{a}$$

$$\frac{\sin B}{5.9} \approx \frac{\sin 38^\circ}{8.1992}$$

$$\sin B \approx 0.443...$$

$$B \approx \sin^{-1}(0.443...)$$

$$\boxed{B \approx 26.2966^\circ}$$

III) Find $m\angle C$.

$$A + B + C = 180^\circ$$

$$38^\circ + 26.2966^\circ + C \approx 180^\circ$$

$$\boxed{C \approx 115.7^\circ}$$

⑤ $c = 4.5 \text{ km}$, $a = 8 \text{ km}$, $B = 128^\circ$

I) Find b .

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$b^2 = 8^2 + 4.5^2 - 2(8)(4.5) \cos 128^\circ$$

$$b^2 = 64 + 20.25 - 72 \cos 128^\circ$$

$$b^2 \approx 84.25 - 72(-0.61566\dots)$$

$$b^2 \approx 84.25 + 44.3276\dots$$

$$b^2 \approx 128.5776\dots$$

$$\boxed{b \approx 11.3392\dots \text{ km}}$$

II) Find $m\angle C$

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin C}{4.5} \approx \frac{\sin 128^\circ}{11.3392\dots}$$

$$\sin C \approx \frac{4.5 \sin 128^\circ}{11.3392\dots}$$

$$\sin C \approx 0.3127\dots$$

$$C \approx \sin^{-1}(0.3127\dots)$$

$$\boxed{C \approx 18.2235\dots}$$

III) Find $m\angle A$.

$$A + B + C = 180^\circ$$

$$A + 128^\circ + 18.2235^\circ \approx 180^\circ$$

$$\boxed{A \approx 33.78^\circ}$$

⑥ $b = 23 \text{ mi}$, $C = 142^\circ$, $a = 18 \text{ mi}$

I) Find c .

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 18^2 + 23^2 - 2(18)(23) \cos 142^\circ$$

$$c^2 = 324 + 529 - 828 \cos 142^\circ$$

$$c^2 \approx 853 - 828(-0.788\dots)$$

$$c^2 \approx 853 + 652.4729\dots$$

$$c^2 \approx 1,505.472\dots$$

$$\boxed{c \approx 38.8004\dots \text{ mi}}$$

II) Find $m\angle A$.

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin A}{18} \approx \frac{\sin 142^\circ}{38.8\dots}$$

$$\sin A \approx \frac{18 \sin 142^\circ}{38.8\dots}$$

$$\sin A \approx 0.2856\dots$$

$$A \approx \sin^{-1}(0.2856\dots)$$

$$\boxed{A \approx 16.5954\dots}$$

III) Find $m\angle B$.

$$A + B + C = 180^\circ$$

$$16.5954^\circ + B + 142^\circ \approx 180^\circ$$

$$\boxed{B \approx 21.4^\circ}$$

The Side-Side-Side Scenario (SSS)

Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle.

• ⑦ $a = 4.1\text{m}$, $b = 3\text{m}$, $c = 2.1\text{m}$

I) Find $m\angle A$.

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$4.1^2 = 3^2 + 2.1^2 - 2(3)(2.1) \cos A$$

$$16.81 = 9 + 4.41 - 12.6 \cos A$$

$$16.81 = 13.41 - 12.6 \cos A$$

$$16.81 - 13.41 = 13.41 - 12.6 \cos A - 13.41$$

$$3.4 = -12.6 \cos A$$

$$\frac{3.4}{-12.6} = \frac{-12.6 \cos A}{-12.6}$$

$$-0.2698... \approx \cos A$$

$$A \approx \cos^{-1}(-0.2698)$$

$$\boxed{A \approx 105.6548^\circ}$$

II) Find $m\angle C$.

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 105.6548^\circ}{4.1} \approx \frac{\sin C}{2.1}$$

$$0.4931... \approx \sin C$$

$$C \approx \sin^{-1}(0.4931...)$$

$$\boxed{C \approx 29.5508^\circ}$$

III) Find $m\angle B$.

$$A + B + C = 180^\circ$$

$$105.6548^\circ + B + 29.5508^\circ \approx 180^\circ$$

$$\boxed{B \approx 44.79^\circ}$$

⑧ $b = 3\text{cm}$, $a = 2\text{cm}$, $c = 2.4\text{cm}$

I) Find $m\angle B$.

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$3^2 = 2^2 + 2.4^2 - 2(2)(2.4) \cos B$$

$$9 = 4 + 5.76 - 9.6 \cos B$$

$$9 = 9.76 - 9.6 \cos B$$

$$9 - 9.76 = 9.76 - 9.6 \cos B - 9.76$$

$$-0.76 = -9.6 \cos B$$

$$\frac{-0.76}{-9.6} = \frac{-9.6 \cos B}{-9.6}$$

$$0.07916... \approx \cos B$$

$$B \approx \cos^{-1}(0.07916...)$$

$$\boxed{B \approx 85.4593^\circ}$$

II) Find $m\angle A$.

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{2} \approx \frac{\sin 85.4593^\circ}{3}$$

$$\sin A \approx 0.66457...$$

$$A \approx \sin^{-1}(0.66457...)$$

$$\boxed{A \approx 41.6496^\circ}$$

III) Find $m\angle C$.

$$A + B + C = 180^\circ$$

$$41.6496^\circ + 85.4593^\circ + C \approx 180^\circ$$

$$\boxed{C \approx 52.89^\circ}$$

• ⑨ $b = 14 \text{ mi}$, $c = 20 \text{ mi}$, $a = 10 \text{ mi}$

I) Find $m\angle C$.

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$20^2 = 10^2 + 14^2 - 2(10)(14) \cos C$$

$$400 = 100 + 196 - 280 \cos C$$

$$400 = 296 - 280 \cos C$$

$$104 = -280 \cos C$$

$$-0.3714... \approx \cos C$$

$$C \approx \cos^{-1}(-0.3714)$$

$$\boxed{C \approx 111.8037^\circ}$$

II) Find $m\angle A$.

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin A}{10} \approx \frac{\sin 111.8037^\circ}{20}$$

$$\sin A \approx 0.46423...$$

$$A \approx \sin^{-1}(0.46423...)$$

$$\boxed{A \approx 27.66^\circ}$$

III) Find $m\angle B$.

$$A + B + C = 180^\circ$$

$$27.66^\circ + B + 111.8037^\circ \approx 180^\circ$$

$$\boxed{B \approx 40.5^\circ}$$

⑩ $b = 2.4 \text{ in}$, $a = 2.2 \text{ in}$, $c = 1.4 \text{ in}$

I) Find $m\angle B$.

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$2.4^2 = 2.2^2 + 1.4^2 - 2(2.2)(1.4) \cos B$$

$$5.76 = 4.84 + 1.96 - 6.16 \cos B$$

$$5.76 = 6.8 - 6.16 \cos B$$

$$-1.04 = -6.16 \cos B$$

$$0.1688... \approx \cos B$$

$$B \approx \cos^{-1}(0.1688...)$$

$$\boxed{B \approx 80.2801^\circ}$$

II) Find $m\angle C$.

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin C}{1.4} \approx \frac{\sin 80.2801^\circ}{2.4}$$

$$\sin C \approx 0.5749...$$

$$C \approx \sin^{-1}(0.5749...)$$

$$\boxed{C \approx 35.0968^\circ}$$

III) Find $m\angle A$.

$$A + B + C = 180^\circ$$

$$A + 80.2801 + 35.0968^\circ \approx 180^\circ$$

$$\boxed{A \approx 64.623^\circ}$$